

Sprint 1- STP Configuration Evidence

Note: This document was created to house multiple screenshots instead of adding each one to the evidence folder for ease of reference

Purpose:

Spanning Tree Protocol (STP) was configured to prevent Layer 2 switching loops caused by redundant connections between the access switches (KFP-SW1 through KFP-SW4) and multilayer switches (KFP-MLS1 & KFP-MLS2). Intentional root bridge placement was used so that the network does not rely on the default STP election process.

Design:

Rapid PVST+ was configured across switches. KFP-MLS1 was selected as the primary root for VLANs 10 (IT/Admin), 20 (Finance), and 99 (Management), while KFP-MLS2 was selected as the primary root for VLANs 30 (General Staff), 40 (Servers), and 50 (Wireless). The opposite multilayer switch was configured to serve as the secondary root for each group.

This placement distributes the Layer 2 topology between the two multilayer switches and aligns the STP roots with the planned gateway/SVI placement for the corresponding VLANs.

Captured from KFP-MLS1:

```
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    24586
           Address    000A.41CD.7A3A
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    24586 (priority 24576 sys-id-ext 10)
           Address    000A.41CD.7A3A
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Desg FWD 19      128.1    P2p
Gi0/2       Desg FWD 4       128.26   P2p
```

```
VLAN0020
Spanning tree enabled protocol rstp
Root ID    Priority    24596
           Address    000A.41CD.7A3A
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    24596 (priority 24576 sys-id-ext 20)
           Address    000A.41CD.7A3A
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/2       Desg FWD 19      128.2    P2p
Gi0/2       Desg FWD 4       128.26   P2p
```

```

VLAN0030
Spanning tree enabled protocol rstp
Root ID    Priority    24606
           Address    0030.F21A.47D6
           Cost        4
           Port        26(GigabitEthernet0/2)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    28702 (priority 28672 sys-id-ext 30)
           Address    000A.41CD.7A3A
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/3          Desg FWD 19      128.3   P2p
Gi0/2          Root FWD 4        128.26  P2p

```

```

VLAN0040
Spanning tree enabled protocol rstp
Root ID    Priority    24616
           Address    0030.F21A.47D6
           Cost        4
           Port        26(GigabitEthernet0/2)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    28712 (priority 28672 sys-id-ext 40)
           Address    000A.41CD.7A3A
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/4          Desg FWD 19      128.4   P2p
Gi0/2          Root FWD 4        128.26  P2p

```

```

VLAN0050
Spanning tree enabled protocol rstp
Root ID    Priority    24626
           Address    0030.F21A.47D6
           Cost        4
           Port        26(GigabitEthernet0/2)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    28722 (priority 28672 sys-id-ext 50)
           Address    000A.41CD.7A3A
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/4          Desg FWD 19      128.4   P2p
Gi0/2          Root FWD 4        128.26  P2p

```

```

VLAN0099
Spanning tree enabled protocol rstp
Root ID    Priority    24675
           Address    000A.41CD.7A3A
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    24675 (priority 24576 sys-id-ext 99)
           Address    000A.41CD.7A3A
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19      128.1   P2p
Fa0/2          Desg FWD 19      128.2   P2p
Fa0/3          Desg FWD 19      128.3   P2p
Fa0/4          Desg FWD 19      128.4   P2p
Gi0/2          Desg FWD 4        128.26  P2p

```

KFP-MLS1>

Captured from KFP-MLS2:

```
VLAN0010
Spanning tree enabled protocol rstp
Root ID    Priority    24586
           Address    000A.41CD.7A3A
           Cost        4
           Port        26(GigabitEthernet0/2)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    28682 (priority 28672 sys-id-ext 10)
           Address    0030.F21A.47D6
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface   Role Sts Cost        Prio.Nbr Type
-----
Fa0/1       Desg FWD 19         128.1   P2p
Gi0/2       Root FWD 4          128.26  P2p
```

```
VLAN0020
Spanning tree enabled protocol rstp
Root ID    Priority    24596
           Address    000A.41CD.7A3A
           Cost        4
           Port        26(GigabitEthernet0/2)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    28692 (priority 28672 sys-id-ext 20)
           Address    0030.F21A.47D6
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface   Role Sts Cost        Prio.Nbr Type
-----
Fa0/2       Desg FWD 19         128.2   P2p
Gi0/2       Root FWD 4          128.26  P2p
```

```
VLAN0030
Spanning tree enabled protocol rstp
Root ID    Priority    24606
           Address    0030.F21A.47D6
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    24606 (priority 24576 sys-id-ext 30)
           Address    0030.F21A.47D6
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface   Role Sts Cost        Prio.Nbr Type
-----
Fa0/3       Desg FWD 19         128.3   P2p
Gi0/2       Desg FWD 4          128.26  P2p
```

```
VLAN0040
Spanning tree enabled protocol rstp
Root ID    Priority    24616
           Address    0030.F21A.47D6
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    24616 (priority 24576 sys-id-ext 40)
           Address    0030.F21A.47D6
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

Interface   Role Sts Cost        Prio.Nbr Type
-----
Fa0/4       Desg FWD 19         128.4   P2p
Gi0/2       Desg FWD 4          128.26  P2p
```

VLAN0050

Spanning tree enabled protocol rstp
Root ID Priority 24626
 Address 0030.F21A.47D6
 This bridge is the root
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24626 (priority 24576 sys-id-ext 50)
 Address 0030.F21A.47D6
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/4	Desg	FWD	19	128.4	P2p
Gi0/2	Desg	FWD	4	128.26	P2p

VLAN0099

Spanning tree enabled protocol rstp
Root ID Priority 24675
 Address 000A.41CD.7A3A
 Cost 4
 Port 26 (GigabitEthernet0/2)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 28771 (priority 28672 sys-id-ext 99)
 Address 0030.F21A.47D6
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Desg	FWD	19	128.3	P2p
Fa0/4	Desg	FWD	19	128.4	P2p
Gi0/2	Root	FWD	4	128.26	P2p